

Polynomial Root-Hitting Bounds from Endpoint Conditional Anti-Concentration

1 Theoretical Setup

1.1 Problem setting

Fix a source parameter domain $\Theta \subseteq \mathbb{R}$, and let

$$\mathcal{I}(\Theta) := \{I \subseteq \Theta : I \text{ is an interval and } 0 < |I| < \infty\},$$

where $|I|$ denotes one-dimensional Lebesgue length and every endpoint convention is allowed. Fix $\eta = (\bar{\kappa}_0, \bar{\kappa}_\infty) \in [0, \infty)^2$. For an integer $d \geq 1$, a radius $R \geq 1$, and $\alpha = (\alpha_0, \dots, \alpha_{d-1}) \in [-R, R]^d$, define the monic polynomial

$$\phi_\alpha(\theta) := \theta^d + \sum_{j=0}^{d-1} \alpha_j \theta^j$$

and the interval root-hitting event

$$H_{d,I} := \{\alpha \in [-R, R]^d : \text{there is } \theta \in I \text{ with } \phi_\alpha(\theta) = 0\}.$$

For a Borel probability law μ supported on $[-R, R]^d$, whenever the indicated regular conditional laws have Lebesgue densities, set

$$\begin{aligned} K_0^\mu &:= \left\| f_{\alpha_0|\alpha_{1:d-1}}^\mu(\cdot | \alpha_{1:d-1}) \right\|_{L^\infty(\mathbb{R})}, \\ K_\infty^\mu &:= \left\| f_{\alpha_{d-1}|\alpha_{0:d-2}}^\mu(\cdot | \alpha_{0:d-2}) \right\|_{L^\infty(\mathbb{R})}. \end{aligned}$$

These are random essential-supremum slice caps, evaluated at the random conditioning coordinates. When $d = 1$, both conditioning tuples are empty and both symbols denote the L^∞ norm of a density of the sole coefficient α_0 .

Writing $\mathcal{P}(\mathbb{R}^d)$ for the Borel probability laws on $\mathbb{R}^d$, define

$$\begin{aligned} \mathcal{D}_{d,R,\eta} &:= \{\mu \in \mathcal{P}(\mathbb{R}^d) : \mu([-R, R]^d) = 1; \\ &\quad \mu(\alpha_0 \in \cdot | \alpha_{1:d-1}) \text{ and } \mu(\alpha_{d-1} \in \cdot | \alpha_{0:d-2}) \text{ admit Lebesgue densities a.s.}; \\ &\quad \mathbb{E}_\mu K_0^\mu \leq \bar{\kappa}_0, \quad \mathbb{E}_\mu K_\infty^\mu \leq \bar{\kappa}_\infty\}. \end{aligned}$$

This definition imposes two mean conditional-density caps. It imposes no almost-sure uniform slice cap, ambient joint density, independence, affine-latent representation, root statistic, or prescribed value of the root-hitting constant. That constant is

$$C_{\mathcal{D}_{d,R,\eta}} := \sup_{\mu \in \mathcal{D}_{d,R,\eta}} \sup_{I \in \mathcal{I}(\Theta)} \frac{\mu(H_{d,I})}{|I|},$$

with value zero when either displayed indexing set is empty.

For a separate nonemptiness statement, define a witness law $\mu_{d,R}^{\text{wit}}$ as follows. If $d = 1$, take $\alpha_0 \sim \text{Unif}[-R, R]$. If $d = 2$, take α_0 and α_1 independently uniform on $[-R, R]$. If $d \geq 3$, let $U_0, U_\infty \sim \text{Unif}[-R, R]$ and let S be Rademacher, all mutually independent, and set

$$\alpha_0 = U_0, \quad \alpha_{d-1} = U_\infty, \quad \alpha_j = RS \quad (1 \leq j \leq d-2).$$

Every witness is supported on $[-R, R]^d$. The proof below also verifies that it is ambiently singular for $d \geq 3$ and has dependent middle coordinates for $d \geq 4$.

1.2 Assumptions

Assumption 1 (Compact parameter domain). $\Theta \subseteq \mathbb{R}$ is a fixed compact source parameter domain. The result is uniform over every $I \in \mathcal{I}(\Theta)$, and its constants do not depend on I or on Θ .

Assumption 2 (Indexed regime). $d \in \mathbb{N}$ with $d \geq 1$, $R \geq 1$, and the finite pair $\eta = (\bar{\kappa}_0, \bar{\kappa}_\infty) \in [0, \infty)^2$ is fixed independently of d and R . No lower bound on either cap parameter is assumed for the root-hitting estimate; the threshold $1/2$ is used only in the separate witness clause.

Assumption 3 (Compact cube support and monicity). Every theorem-quantified law μ satisfies $\mu([-R, R]^d) = 1$, and its polynomial is exactly the monic ϕ_α above. This is only a support condition and does not impose an ambient density.

Assumption 4 (Mean endpoint conditional-density caps). The two endpoint regular conditional laws admit Lebesgue densities almost surely, and their random essential-supremum norms satisfy exactly

$$\mathbb{E}_\mu K_0^\mu \leq \bar{\kappa}_0, \quad \mathbb{E}_\mu K_\infty^\mu \leq \bar{\kappa}_\infty.$$

These are mean bounds only. Together with Assumption 3, they exhaust the membership requirements defining $\mathcal{D}_{d,R,\eta}$.

1.3 Objective and scope

The objective is an explicit, class-uniform polynomial upper bound for $C_{\mathcal{D}_{d,R,\eta}}$, together with the sharper interval-piece bound from which it follows. The coefficients may depend on the fixed η but on no other hidden quantity. Under the separate condition $\bar{\kappa}_0, \bar{\kappa}_\infty \geq 1/2$, the witness above should certify nonemptiness for every $d \geq 1$ and $R \geq 1$.

This is a material partial result for the source problem. It gives a sufficient coefficient-side condition and an explicit $O_\eta(Rd^2)$ rate throughout the restricted regime $R \geq 1$. It neither claims necessity nor covers laws without both endpoint mean caps or the regime $0 < R < 1$. No learner or online algorithm is specialized.

2 Preliminaries

For $I \in \mathcal{I}(\Theta)$, introduce the three setting-derived chart pieces

$$I_0 := I \cap [-1, 1], \quad I_+ := I \cap (1, \infty), \quad I_- := I \cap (-\infty, -1).$$

They record the exact interval lengths that appear in the main theorem. The corresponding coefficient sweeps are

$$s_0(\theta; \alpha_{1:d-1}) := -\theta^d - \sum_{j=1}^{d-1} \alpha_j \theta^j, \quad |\theta| \leq 1,$$

$$s_\infty(\theta; \alpha_{0:d-2}) := -\theta - \sum_{j=0}^{d-2} \alpha_j \theta^{j-d+1}, \quad |\theta| \geq 1.$$

All empty sums are zero. These maps express the root equation using α_0 on the inner chart and α_{d-1} on the two outer charts.

The exact chart coefficients and their fixed- η envelope are

$$B_0(d, R) := d + \frac{Rd(d-1)}{2},$$

$$B_\infty(d, R) := 1 + \frac{Rd(d-1)}{2},$$

$$M_\eta(d, R) := \max\{\bar{\kappa}_0 B_0(d, R), \bar{\kappa}_\infty B_\infty(d, R)\},$$

$$\bar{\kappa}_* := \max\{\bar{\kappa}_0, \bar{\kappa}_\infty\},$$

$$P_\eta(d, R) := \bar{\kappa}_* d + \frac{\bar{\kappa}_*}{2} Rd^2.$$

The main theorem uses B_0 and B_∞ for its sharp weighted estimate, M_η for the exact two-chart maximum, and P_η for the public polynomial specialization. For a scalar map s and a set J , the notation $s(J) := \{s(\theta) : \theta \in J\}$ denotes its direct image.

3 Main Theorem

Theorem 3.1 (Endpoint conditional anti-concentration). *Under Assumptions 1–4, fix the finite $\eta = (\bar{\kappa}_0, \bar{\kappa}_\infty)$. For every $d \geq 1$, $R \geq 1$, $\mu \in \mathcal{D}_{d,R,\eta}$, and $I \in \mathcal{I}(\Theta)$,*

$$\begin{aligned} \mu(H_{d,I}) &\leq \bar{\kappa}_0 B_0(d, R) |I_0| + \bar{\kappa}_\infty B_\infty(d, R) (|I_+| + |I_-|) \\ &\leq M_\eta(d, R) |I|. \end{aligned}$$

Consequently,

$$C_{\mathcal{D}_{d,R,\eta}} \leq M_\eta(d, R) \leq P_\eta(d, R) = \bar{\kappa}_* d + \frac{\bar{\kappa}_*}{2} Rd^2.$$

The middle quantity is the maximum of the two chart coefficients, rather than their sum.

Separately, if $\bar{\kappa}_0, \bar{\kappa}_\infty \geq 1/2$, then for every $d \geq 1$ and $R \geq 1$ the law $\mu_{d,R}^{\text{wit}}$ belongs to $\mathcal{D}_{d,R,\eta}$, and

$$K_0^{\mu_{d,R}^{\text{wit}}} = K_\infty^{\mu_{d,R}^{\text{wit}}} = \frac{1}{2R} \quad \text{almost surely.}$$

This threshold is not required for either root-hitting inequality.

All exposed dependence is on d , R , $\bar{\kappa}_0$, $\bar{\kappa}_\infty$, and the displayed interval lengths; there are no hidden constants. The pair η and the domain Θ are fixed, and the bound itself is independent of Θ . The statement is a deterministic inequality for each probability $\mu(H_{d,I})$, followed by deterministic suprema, with no confidence parameter. It is static and uniform over all positive-length intervals, with no horizon or limiting qualification. Its norm and metric inputs are the conditional $L^\infty(\mathbb{R})$ density norms and one-dimensional Lebesgue interval length.

Corollary 3.1 (Fixed-cap polynomial rate). *Under Assumptions 1–4, for the single fixed finite η and every $d \geq 1$, $R \geq 1$,*

$$C_{\mathcal{D}_{d,R,\eta}} \leq \bar{\kappa}_* d + \frac{\bar{\kappa}_*}{2} R d^2.$$

This is a pointwise, deterministic, static bound with no hidden constant or probability conversion; its displayed polynomial has total degree at most three in (d, R) and coefficients depending only on η .

Proof. Theorem 3.1 gives $C_{\mathcal{D}_{d,R,\eta}} \leq M_\eta(d, R)$. The complete scalar specialization, including the choices $\bar{\kappa}_* = \max\{\bar{\kappa}_0, \bar{\kappa}_\infty\}$, the conditions $d \geq 1$ and $R \geq 1$, the inequalities $1 \leq d$ and $d(d-1) \leq d^2$, and the absence of any probability-mode conversion, is proved in Proposition A.10. Combining the two statements gives the displayed rate. $\square$

4 Proof Sketch

The proof uses two coefficient pivots. On $[-1, 1]$, solve the root equation for α_0 . Lemma A.1 bounds the derivative of the resulting sweep, and Lemma A.2 converts this into an image-length estimate. On each connected outer component, divide by θ^{d-1} and solve for α_{d-1} . Propositions A.1 and A.2 establish the same certificate separately on the two signs, without crossing zero or assigning ± 1 to an outer chart.

The probabilistic interface requires some care because the hypotheses bound only the means of random conditional-density caps. Lemma A.3 constructs measurable, version-independent representatives of those caps and proves simultaneous domination of all Borel pivot sets. Lemma A.4 proves that the root events are Borel and identifies their exact conditional sections. Proposition A.3 then matches each section to the correct regular conditional kernel.

Propositions A.4, A.5, and A.6 integrate the deterministic image bounds against their matching random caps. Lemma A.5 partitions the parameter interval exactly, and Propositions A.7 and A.8 combine the three event controls into the weighted estimate and then the exact maximum. Finally, Proposition A.9 takes the two suprema, while Proposition A.10 gives the explicit polynomial envelope.

The nonemptiness clause is independent of the root-hitting argument. Three degree-specific kernel propositions compute both witness caps as $1/(2R)$. The remaining witness propositions verify cube support, ambient singularity, and middle-coordinate dependence, and Lemma A.8 isolates the only use of the threshold $1/2$. The final appendix proof assembles these results into Theorem 3.1.

A Proof Details

A.1 Inner derivative and image length

Lemma A.1 (Inner-sweep derivative and Lipschitz control). *Under Assumptions 2 and 3, if $d \geq 1$, $R \geq 1$, and $\alpha_{1:d-1} \in [-R, R]^{d-1}$, then for every $\theta \in [-1, 1]$,*

$$\partial_\theta s_0(\theta; \alpha_{1:d-1}) = -d\theta^{d-1} - \sum_{j=1}^{d-1} j\alpha_j\theta^{j-1}$$

and

$$|\partial_\theta s_0(\theta; \alpha_{1:d-1})| \leq d + \frac{Rd(d-1)}{2} = B_0(d, R).$$

Consequently, for all $x, z \in [-1, 1]$,

$$|s_0(x; \alpha_{1:d-1}) - s_0(z; \alpha_{1:d-1})| \leq B_0(d, R)|x - z|.$$

Proof. The definition

$$s_0(\theta; \alpha_{1:d-1}) = -\theta^d - \sum_{j=1}^{d-1} \alpha_j \theta^j$$

is polynomial in θ , so termwise differentiation gives the asserted formula on all of $\mathbb{R}$. First suppose $d \geq 2$. For $\theta \in [-1, 1]$, Assumption 3 and the triangle inequality give

$$\begin{aligned} |\partial_\theta s_0(\theta; \alpha_{1:d-1})| &\leq d|\theta|^{d-1} + \sum_{j=1}^{d-1} j|\alpha_j||\theta|^{j-1} \\ &\leq d + R \sum_{j=1}^{d-1} j \\ &= d + \frac{Rd(d-1)}{2} = B_0(d, R). \end{aligned}$$

No term has been absorbed. For distinct $x, z \in [-1, 1]$, the closed segment between them lies in $[-1, 1]$; the mean-value theorem therefore yields the Lipschitz inequality. It is immediate when $x = z$, and the argument includes both endpoints -1 and 1 .

When $d = 1$, the coefficient tuple and every displayed sum are empty. Without using a 0^0 convention,

$$s_0(\theta) = -\theta, \quad s'_0(\theta) = -1, \quad B_0(1, R) = 1,$$

so the derivative and Lipschitz conclusions remain exact. At the radius boundary $R = 1$, the same computation uses $|\alpha_j| \leq 1$ and gives $B_0(d, 1) = d + d(d-1)/2$; no strict radius inequality or limiting argument is needed. $\square$

Lemma A.2 (Inner interval image length). *Under Assumptions 2 and 3 and Lemma A.1, if $d \geq 1$, $R \geq 1$, $\alpha_{1:d-1} \in [-R, R]^{d-1}$, and $J \subseteq [-1, 1]$ is an interval with any endpoint convention, then $s_0(J; \alpha_{1:d-1})$ is an interval or is empty, and*

$$\lambda(s_0(J; \alpha_{1:d-1})) \leq B_0(d, R)|J|.$$

This includes empty and singleton J , intervals containing either boundary point, and $d = 1$.

Proof. If $J = \emptyset$, its image is empty and both lengths are zero. If $J = \{\theta_0\}$, its image is a singleton and again both lengths are zero. Suppose now that J has at least two points. Every real interval is connected, regardless of endpoint inclusion. Since s_0 is continuous, its image is connected and hence is an interval in $\mathbb{R}$. It is bounded because Lemma A.1 makes s_0 Lipschitz on the bounded set $[-1, 1]$.

For arbitrary u, v in the image, choose $x, z \in J$ with $u = s_0(x; \alpha_{1:d-1})$ and $v = s_0(z; \alpha_{1:d-1})$. Then

$$|u - v| \leq B_0(d, R)|x - z| \leq B_0(d, R) \text{diam}(J).$$

Taking the supremum over u, v gives

$$\text{diam}(s_0(J; \alpha_{1:d-1})) \leq B_0(d, R) \text{diam}(J).$$

The Lebesgue length of any nonempty bounded real interval equals its diameter, independently of whether either endpoint is included. Hence

$$\begin{aligned} \lambda(s_0(J; \alpha_{1:d-1})) &= \text{diam}(s_0(J; \alpha_{1:d-1})) \\ &\leq B_0(d, R) \text{diam}(J) = B_0(d, R)|J|. \end{aligned}$$

This proves the complete inner deterministic certificate. No monotonicity, injectivity, transversality, or simple-root condition is used. In degree one the image is exactly $-J$, so equality holds with $B_0(1, R) = 1$. $\square$

A.2 Positive and negative outer certificates

Proposition A.1 (Positive outer pivot and image certificate). *Under Assumptions 2 and 3, fix $\alpha \in [-R, R]^d$. If $J \subseteq (1, \infty)$ is an interval with any endpoint convention, including an empty or singleton interval, then for every $\theta \in J$,*

$$\phi_\alpha(\theta) = 0 \iff \alpha_{d-1} = s_\infty(\theta; \alpha_{0:d-2}) = -\theta - \sum_{j=0}^{d-2} \alpha_j \theta^{j-d+1}.$$

Moreover,

$$\partial_\theta s_\infty(\theta; \alpha_{0:d-2}) = -1 + \sum_{j=0}^{d-2} (d-1-j) \alpha_j \theta^{j-d},$$

$$|\partial_\theta s_\infty(\theta; \alpha_{0:d-2})| \leq B_\infty(d, R), \quad B_\infty(d, R) = 1 + \frac{Rd(d-1)}{2},$$

and

$$\lambda(s_\infty(J; \alpha_{0:d-2})) \leq B_\infty(d, R)|J|.$$

The conclusions remain valid as an endpoint approaches 1 from above, and for $d = 1$ they reduce to $s_\infty = -\theta$ and exact image length $|J|$.

Proof. For $\theta > 1$, the divisor θ^{d-1} is nonzero. Separating the coefficient of degree $d-1$ gives the reversible chain

$$\begin{aligned} \phi_\alpha(\theta) = 0 &\iff \frac{\phi_\alpha(\theta)}{\theta^{d-1}} = 0 \\ &\iff \theta + \sum_{j=0}^{d-2} \alpha_j \theta^{j-d+1} + \alpha_{d-1} = 0 \\ &\iff \alpha_{d-1} = -\theta - \sum_{j=0}^{d-2} \alpha_j \theta^{j-d+1}. \end{aligned}$$

Termwise differentiation yields

$$\begin{aligned}\partial_\theta s_\infty(\theta; \alpha_{0:d-2}) &= -1 - \sum_{j=0}^{d-2} (j-d+1) \alpha_j \theta^{j-d} \\ &= -1 + \sum_{j=0}^{d-2} (d-1-j) \alpha_j \theta^{j-d}.\end{aligned}$$

For $d \geq 2$, every exponent in the last sum satisfies $j-d \leq -2$ and $|\theta^{j-d}| \leq 1$. Therefore

$$\begin{aligned}|\partial_\theta s_\infty(\theta; \alpha_{0:d-2})| &\leq 1 + \sum_{j=0}^{d-2} (d-1-j) |\alpha_j| |\theta^{j-d}| \\ &\leq 1 + R \sum_{j=0}^{d-2} (d-1-j) \\ &= 1 + R \sum_{k=1}^{d-1} k = 1 + \frac{Rd(d-1)}{2} = B_\infty(d, R).\end{aligned}$$

The final identity also covers $d = 1$, when both sums are empty.

If $|J| = \infty$, the image-length inequality holds in the extended nonnegative reals. If J is empty or a singleton, both sides have length zero. Otherwise, every segment joining two points of J remains in $(1, \infty)$. The mean-value theorem gives the $B_\infty(d, R)$ -Lipschitz bound there. The continuous image of the connected set J is an interval, and the length-diameter identity for real intervals gives

$$\begin{aligned}\lambda(s_\infty(J; \alpha_{0:d-2})) &= \text{diam}(s_\infty(J; \alpha_{0:d-2})) \\ &\leq B_\infty(d, R) \text{diam}(J) = B_\infty(d, R)|J|.\end{aligned}$$

This is unchanged by inclusion or exclusion of a finite endpoint.

The one-sided limits

$$\begin{aligned}\lim_{\theta \downarrow 1} s_\infty(\theta; \alpha_{0:d-2}) &= -1 - \sum_{j=0}^{d-2} \alpha_j, \\ \lim_{\theta \downarrow 1} \partial_\theta s_\infty(\theta; \alpha_{0:d-2}) &= -1 + \sum_{j=0}^{d-2} (d-1-j) \alpha_j\end{aligned}$$

are finite and satisfy the same derivative bound. Thus approaching the excluded endpoint adds no length term; the point 1 itself belongs to the inner chart. When $d = 1$, division is by $\theta^0 = 1$, all sums are empty, $\phi_\alpha(\theta) = \theta + \alpha_0$, and all stated identities are exact. $\square$

Proposition A.2 (Negative outer pivot and image certificate). *Under Assumptions 2 and 3, fix $\alpha \in [-R, R]^d$. If $J \subseteq (-\infty, -1)$ is an interval with any endpoint convention, including an empty or singleton interval, then for every $\theta \in J$,*

$$\phi_\alpha(\theta) = 0 \iff \alpha_{d-1} = s_\infty(\theta; \alpha_{0:d-2}) = -\theta - \sum_{j=0}^{d-2} \alpha_j \theta^{j-d+1}.$$

The exact derivative is

$$\partial_\theta s_\infty(\theta; \alpha_{0:d-2}) = -1 + \sum_{j=0}^{d-2} (d-1-j) \alpha_j \theta^{j-d},$$

and it satisfies

$$|\partial_\theta s_\infty(\theta; \alpha_{0:d-2})| \leq B_\infty(d, R), \quad \lambda(s_\infty(J; \alpha_{0:d-2})) \leq B_\infty(d, R)|J|.$$

These conclusions preserve the negative- θ parity signs, remain valid as an endpoint approaches -1 from below, and specialize exactly at $d = 1$.

Proof. For $\theta < -1$, the number θ^{d-1} is nonzero. It may be negative, but it divides an equality rather than an inequality, so

$$\begin{aligned} \phi_\alpha(\theta) = 0 &\iff \frac{\phi_\alpha(\theta)}{\theta^{d-1}} = 0 \\ &\iff \theta + \sum_{j=0}^{d-2} \alpha_j \theta^{j-d+1} + \alpha_{d-1} = 0 \\ &\iff \alpha_{d-1} = -\theta - \sum_{j=0}^{d-2} \alpha_j \theta^{j-d+1}. \end{aligned}$$

Multiplication by the same nonzero divisor proves the converse. Direct differentiation gives the derivative displayed in the proposition.

To retain every parity sign, orient this connected chart by $u = -\theta > 1$ and put

$$g(u) := s_\infty(-u; \alpha_{0:d-2}) = u - \sum_{j=0}^{d-2} \alpha_j (-u)^{j-d+1}.$$

The chain rule gives

$$\begin{aligned} g'(u) &= -\partial_\theta s_\infty(-u; \alpha_{0:d-2}) \\ &= 1 - \sum_{j=0}^{d-2} (d-1-j) \alpha_j (-u)^{j-d} \\ &= 1 - \sum_{j=0}^{d-2} (d-1-j) \alpha_j (-1)^{d-j} u^{j-d}. \end{aligned}$$

Here $(-u)^{j-d} = (-1)^{j-d} u^{j-d} = (-1)^{d-j} u^{j-d}$ because the exponents are integers. Thus orientation reversal gives exactly $g'(u) = -s'_\infty(-u)$ and assigns no favorable sign to a coefficient term.

For $d \geq 2$, $j-d \leq -2$ and $|\theta^{j-d}| = |\theta|^{-(d-j)} \leq 1$. Consequently,

$$\begin{aligned} |\partial_\theta s_\infty(\theta; \alpha_{0:d-2})| &\leq 1 + \sum_{j=0}^{d-2} (d-1-j) |\alpha_j| |\theta^{j-d}| \\ &\leq 1 + R \sum_{j=0}^{d-2} (d-1-j) \\ &= 1 + R \sum_{k=1}^{d-1} k = 1 + \frac{Rd(d-1)}{2} = B_\infty(d, R). \end{aligned}$$

The same bound holds for $|g'(u)|$, and the triangular-sum identity includes the empty $d = 1$ sum.

If $|J| = \infty$, the length conclusion is immediate in the extended reals. Empty and singleton intervals give $0 \leq 0$. Otherwise, every segment joining points of J stays inside $(-\infty, -1)$ and

crosses neither -1 nor zero. The mean-value theorem, connectedness, and the length-diameter identity therefore give

$$\begin{aligned}\lambda(s_\infty(J; \alpha_{0:d-2})) &= \text{diam}(s_\infty(J; \alpha_{0:d-2})) \\ &\leq B_\infty(d, R) \text{diam}(J) = B_\infty(d, R)|J|.\end{aligned}$$

Finally, the relevant one-sided limits are

$$\begin{aligned}\lim_{\theta \uparrow -1} s_\infty(\theta; \alpha_{0:d-2}) &= 1 - \sum_{j=0}^{d-2} \alpha_j (-1)^{j-d+1}, \\ \lim_{\theta \uparrow -1} \partial_\theta s_\infty(\theta; \alpha_{0:d-2}) &= -1 + \sum_{j=0}^{d-2} (d-1-j) \alpha_j (-1)^{j-d}, \\ \lim_{u \downarrow 1} g'(u) &= 1 - \sum_{j=0}^{d-2} (d-1-j) \alpha_j (-1)^{j-d}.\end{aligned}$$

The last two quantities are negatives and each has absolute value at most $B_\infty(d, R)$. Thus approaching -1 introduces neither a blow-up nor an endpoint correction; -1 belongs to the inner chart. In degree one all sums are empty, division is by 1, and $s_\infty = -\theta$ on this chart as well. The two outer propositions have treated the connected components separately and nowhere join points across zero. $\square$

A.3 Measurable caps, Borel sections, and matching disintegration

Lemma A.3 (Measurable endpoint-kernel caps). *Under Assumptions 2, 3, and 4, let $\mu \in \mathcal{D}_{d,R,\eta}$. For $d \geq 2$, define*

$$(X_0, Y_0) := (\alpha_0, \alpha_{1:d-1}), \quad (X_\infty, Y_\infty) := (\alpha_{d-1}, \alpha_{0:d-2}),$$

let ν_i be the law of Y_i , and fix a regular conditional kernel $Q_y^i = \mathcal{L}(X_i \mid Y_i = y)$ for $i \in \{0, \infty\}$. For $d = 1$, take both pairs to be $(\alpha_0, \emptyset)$ and use the same unconditional kernel on $\mathbb{R}^0 = \{\emptyset\}$.

There exist extended nonnegative measurable functions $\hat{K}_i$ and ν_i -full measurable sets G_i such that every $y \in G_i$ is support-compatible and

$$Q_y^i \ll \lambda, \quad Q_y^i([-R, R]) = 1, \quad \hat{K}_i(y) = \left\| \frac{dQ_y^i}{d\lambda} \right\|_{L^\infty(\mathbb{R})} < \infty.$$

On the same set, simultaneously for every Borel $A \subseteq \mathbb{R}$,

$$Q_y^i(A) \leq \hat{K}_i(y) \lambda(A). \tag{A.1}$$

Furthermore,

$$\hat{K}_0(Y_0) = K_0^\mu, \quad \hat{K}_\infty(Y_\infty) = K_\infty^\mu \quad \mu\text{-a.s.}, \tag{A.2}$$

and

$$\mathbb{E}_\mu \hat{K}_0(Y_0) \leq \bar{\kappa}_0, \quad \mathbb{E}_\mu \hat{K}_\infty(Y_\infty) \leq \bar{\kappa}_\infty. \tag{A.3}$$

The representatives are unchanged almost surely if a conditional-kernel or density version is changed. In degree one the two kernels and caps coincide.

Proof. Fix one index i , retaining its own conditioning marginal ν_i and kernel Q^i throughout. The argument will then be applied independently to $i = 0$ and $i = \infty$.

By Assumption 4, $Q_y^i \ll \lambda$ outside a ν_i -null set. Write S_i for the conditioning space, namely $\mathbb{R}^{d-1}$ when $d \geq 2$ and $\mathbb{R}^0$ when $d = 1$. The marginal ν_i is a probability measure on S_i , and Q^i is a probability kernel from S_i to $\mathbb{R}$. The measure–kernel product theorem (Kallenberg, 2002, Chapter 1), applied to these objects, defines on $S_i \times \mathbb{R}$ the measure

$$\gamma_i(C) := \int Q_y^i(C_y) \nu_i(dy), \quad C_y := \{x : (y, x) \in C\}.$$

It supplies the measurability of the section integral and countable additivity. Moreover,

$$\gamma_i(S_i \times \mathbb{R}) = \int Q_y^i(\mathbb{R}) \nu_i(dy) = \int 1 d\nu_i = 1,$$

so γ_i is a probability before the Radon–Nikodym theorem is invoked. For every Borel rectangle $D \times A$, regular conditionality gives

$$\gamma_i(D \times A) = \int_D Q_y^i(A) \nu_i(dy) = \mu(Y_i \in D, X_i \in A).$$

Rectangles form a π -system containing the whole product and generate its product sigma-algebra. The finite measures γ_i and $\mathcal{L}_\mu(Y_i, X_i)$ have the same whole-space mass and agree on every rectangle. Finite-measure uniqueness therefore identifies the current objects exactly:

$$\gamma_i = \mathcal{L}_\mu(Y_i, X_i).$$

If $(\nu_i \otimes \lambda)(C) = 0$, Tonelli’s theorem (Folland, 1999, Chapter 2) gives $\lambda(C_y) = 0$ for ν_i -almost every y . Conditional absolute continuity then gives $Q_y^i(C_y) = 0$, and hence $\gamma_i \ll \nu_i \otimes \lambda$. The ordinary product measure $\beta_i := \nu_i \otimes \lambda$ is sigma-finite before Radon–Nikodym is used: the sets $S_i \times [-n, n]$, $n \geq 1$, cover the whole product and

$$\beta_i(S_i \times [-n, n]) = \nu_i(S_i)\lambda([-n, n]) = 2n < \infty.$$

This remains true when $S_i = \mathbb{R}^0$ and requires no density of the possibly singular marginal ν_i . Since γ_i is finite and $\gamma_i \ll \beta_i$, the Radon–Nikodym theorem (Folland, 1999, Chapter 3) supplies a nonnegative jointly measurable $q_i(y, x)$ such that

$$\gamma_i(dy, dx) = q_i(y, x) \nu_i(dy) \lambda(dx).$$

The whole-space calculation also gives $\int q_i d(\nu_i \otimes \lambda) = 1$. Choose the explicit countable family

$$\mathcal{P}_\mathbb{Q} := \{\emptyset, \mathbb{R}\} \cup \{(a, b] : a, b \in \mathbb{Q}, a < b\} \subseteq \mathcal{B}(\mathbb{R}),$$

which is a π -system, contains the whole pivot space, and generates $\mathcal{B}(\mathbb{R})$. Intersections remain in the family or become empty, and every open interval is a countable union of rational half-open members. For every $B \in \mathcal{P}_\mathbb{Q}$ and Borel $D \subseteq S_i$, comparison on $D \times B$ gives

$$\int_D Q_y^i(B) \nu_i(dy) = \int_D \left(\int_B q_i(y, x) dx \right) \nu_i(dy).$$

Both integrands are nonnegative and integrable. Equality on every D implies equality almost everywhere. Intersect the resulting full-measure sets over the countable $\mathcal{P}_\mathbb{Q}$. On that common set define

$$\tilde{Q}_y^i(B) := \int_B q_i(y, x) dx.$$

The member $B = \mathbb{R}$ gives $Q_y^i(\mathbb{R}) = \tilde{Q}_y^i(\mathbb{R}) = 1$. Thus the two fiber measures are finite, have equal whole-space mass, and agree on the generating π -system. The π - λ theorem and its finite-measure uniqueness consequence (Folland, 1999, Chapter 1) extend their equality to every Borel set. Hence $q_i(y, \cdot)$ is a probability density of Q_y^i on one ν_i -full set.

Let

$$\mathcal{J}_{\mathbb{Q}} := \{(a, b) : a, b \in \mathbb{Q}, a < b\}, \quad \hat{K}_i(y) := \sup_{J \in \mathcal{J}_{\mathbb{Q}}} \frac{Q_y^i(J)}{|J|}.$$

For each fixed J , the kernel property makes $y \mapsto Q_y^i(J)$ measurable. The family is countable, so $\hat{K}_i$ is measurable on the whole conditioning space, including fibers on which no density was chosen.

Fix a density fiber, write its density as q , and set $m = \|q\|_{L^\infty(\mathbb{R})}$, allowing $m = +\infty$. For every rational interval,

$$\frac{Q_y^i(J)}{|J|} = \frac{1}{|J|} \int_J q(x) dx \leq m,$$

so $\hat{K}_i(y) \leq m$. If $m = 0$, nonnegativity gives equality. Otherwise take a finite nonnegative $a < m$. The set $\{x : q(x) > a\}$ has positive measure and therefore contains a Lebesgue point of q by the Lebesgue differentiation theorem (Folland, 1999, Chapter 3). Some bounded open interval L around that point satisfies

$$\frac{1}{|L|} \int_L q(t) dt > a.$$

Put

$$\delta := \int_L q d\lambda - a|L| > 0, \quad \rho(E) := \int_E |q| d\lambda.$$

The measure ρ is finite because q is a probability density. Its continuity on sets of small Lebesgue measure follows from the explicit truncation inequality: for every measurable E and every $N \geq 1$,

$$\rho(E) \leq N\lambda(E) + \int_{\{|q| > N\}} |q| d\lambda.$$

Indeed, split E between $\{|q| \leq N\}$ and its complement. Moreover, $|q|\mathbf{1}_{\{|q| \leq N\}} \uparrow |q|$, so monotone convergence (Folland, 1999, Chapter 2) makes the tail integral tend to zero. Given $\varepsilon > 0$, choose N so that the tail is below $\varepsilon/2$; then $\lambda(E) < \varepsilon/(2N)$ implies $\rho(E) < \varepsilon$.

Choose rational endpoints r_n, s_n tending to the endpoints of L , with $r_n < s_n$, and set $J_n = (r_n, s_n)$. Then $\lambda(L \triangle J_n) \rightarrow 0$ and $|J_n| \rightarrow |L|$. The preceding continuity and the definition of δ give

$$\begin{aligned} \int_{J_n} q d\lambda - a|J_n| &\geq \int_L q d\lambda - a|L| - \rho(L \triangle J_n) - a||J_n| - |L|| \\ &= \delta - \rho(L \triangle J_n) - a||J_n| - |L||. \end{aligned}$$

The last two terms tend to zero. For all sufficiently large n , the right-hand side is positive, so some rational interval has average strictly larger than a and $\hat{K}_i(y) > a$. Letting $a \uparrow m$, or taking arbitrarily large finite a when $m = +\infty$, gives

$$\hat{K}_i(y) = \|q\|_{L^\infty(\mathbb{R})} \tag{A.4}$$

on every density fiber.

Two densities of the same fiber law agree λ -almost everywhere by Radon–Nikodym uniqueness (Folland, 1999, Chapter 3), so their essential suprema agree. If Q^i and $\tilde{Q}^i$ are two regular conditional kernels for the same pair (X_i, Y_i) , then for every Borel $D \subseteq S_i$ and every $J \in \mathcal{J}_{\mathbb{Q}}$,

$$\int_D Q_y^i(J) \nu_i(dy) = \mu(Y_i \in D, X_i \in J) = \int_D \tilde{Q}_y^i(J) \nu_i(dy).$$

Apply this equality to the measurable set on which one of the two bounded integrands is strictly larger than the other, and then reverse their roles. It follows that $Q_y^i(J) = \tilde{Q}_y^i(J)$ almost everywhere. Countability gives one common full-measure set for all rational intervals, and the defining supremum gives the same cap there. This proves both density-version and kernel-version independence without selecting values on null fibers.

On a density fiber with finite cap, $q(x) \leq \hat{K}_i(y)$ for λ -almost every x . Consequently, for every Borel A on that same fiber, with no A -dependent exceptional set,

$$Q_y^i(A) = \int_A q(x) dx \leq \hat{K}_i(y) \lambda(A).$$

Equation (A.4) identifies the random cap with the one in the definition of $\mathcal{D}_{d,R,\eta}$, proving (A.2). The primitive mean inequalities give (A.3) without changing constants. Their finite right sides imply that both caps are finite almost surely.

It remains to put conditional support on the same full-measure set. By Assumption 3,

$$0 = \mu(X_i \notin [-R, R]) = \int Q_y^i(\mathbb{R} \setminus [-R, R]) \nu_i(dy).$$

The nonnegative integrand vanishes almost everywhere, and Y_i lies in its corresponding $(d-1)$ -dimensional cube almost surely. Intersecting the full-measure sets for density, finite cap, and support produces G_i . Applying the construction separately to 0 and ∞ proves the result for $d \geq 2$.

For $d = 1$, the conditioning marginal is the unit mass on $\mathbb{R}^0$. The kernel at its sole point is the unconditional law of α_0 , and the same kernel is used for both endpoint labels. The rational-interval formula therefore produces one cap, and all preceding identities give both required inequalities for it. $\square$

Lemma A.4 (Borel root events and exact chart sections). *Under Assumptions 1, 2, and 3, let $J \subseteq \Theta$ be an interval with any endpoint convention, possibly empty or a singleton. Then $H_{d,J}$ is Borel. For $d \geq 2$, define*

$$T_0(x, y) := (x, y_1, \dots, y_{d-1}), \quad T_\infty(x, y) := (y_0, \dots, y_{d-2}, x),$$

and $E_J^i := T_i^{-1}(H_{d,J})$. For $d = 1$, define both maps by $T_i(x, \emptyset) = (x)$. The sets E_J^i and all their pivot sections $(E_J^i)_y := \{x : (x, y) \in E_J^i\}$ are Borel. If $J \subseteq [-1, 1]$, then

$$(E_J^0)_y = \begin{cases} [-R, R] \cap s_0(J; y), & y \in [-R, R]^{d-1}, \\ \emptyset, & y \notin [-R, R]^{d-1}, \end{cases} \quad (\text{A.5})$$

and if $J \subseteq (1, \infty)$ or $J \subseteq (-\infty, -1)$, then

$$(E_J^\infty)_y = \begin{cases} [-R, R] \cap s_\infty(J; y), & y \in [-R, R]^{d-1}, \\ \emptyset, & y \notin [-R, R]^{d-1}. \end{cases} \quad (\text{A.6})$$

For $d = 1$, both right sides mean $[-R, R] \cap (-J)$.

Proof. Assumption 1 makes J bounded. The empty interval is a countable union of empty compact sets, and a singleton is compact. Otherwise write $a = \inf J < b = \sup J$. For all sufficiently large n , set

$$a_n := \begin{cases} a, & a \in J, \\ a + 1/n, & a \notin J, \end{cases} \quad b_n := \begin{cases} b, & b \in J, \\ b - 1/n, & b \notin J, \end{cases}$$

and let $K_n = [a_n, b_n]$ when $a_n \leq b_n$, taking $K_n = \emptyset$ otherwise. After reindexing, the compact sets K_n increase to J . This construction preserves open, closed, and half-open endpoint conventions.

For every n , define

$$Z_n := \{(\alpha, \theta) \in [-R, R]^d \times K_n : \phi_\alpha(\theta) = 0\}.$$

The polynomial evaluation map is continuous. The finite-product, closed-subset, and continuous-image compactness facts (Munkres, 2000, Sections 24 and 26) show that Z_n is compact and that its coefficient projection is compact and Borel. Exact existential quantification gives

$$H_{d,J} = \bigcup_{n=1}^{\infty} \text{proj}_\alpha Z_n.$$

Thus $H_{d,J}$ is Borel. The proof projects compact sets before taking a countable union; it does not rely on projections of arbitrary Borel sets.

The reconstruction maps are continuous, so each E_J^i is Borel. Sections of product Borel sets are Borel: rectangles have Borel sections, and this property is closed under complements and countable unions.

For an inner interval and support-compatible y ,

$$\phi_{T_0(x,y)}(\theta) = 0 \iff x = -\theta^d - \sum_{j=1}^{d-1} y_j \theta^j = s_0(\theta; y).$$

The coefficient cube then gives (A.5); an incompatible y gives an empty section. On either outer chart, $\theta \neq 0$, and

$$\begin{aligned} \phi_{T_\infty(x,y)}(\theta) = 0 &\iff \theta + \sum_{j=0}^{d-2} y_j \theta^{j-d+1} + x = 0 \\ &\iff x = s_\infty(\theta; y), \end{aligned}$$

where division is only by the nonzero θ^{d-1} . This proves (A.6) on both signs. In degree one both sums are empty and both reconstructions are (x) .

Only the zero-set identity is used, so tangencies and multiple roots need no extra case. Empty and singleton intervals and every endpoint convention were included in the compact exhaustion. In particular, ± 1 occur only in the inner chart when they belong to J . $\square$

Proposition A.3 (Matching-kernel Borel disintegration). *Under Assumptions 1, 2, 3, and 4, and under Lemmas A.3 and A.4, use the pairs, kernels, marginals, full-measure sets, and reconstruction events defined there. For every inner-chart interval $J \subseteq \Theta \cap [-1, 1]$,*

$$\mu(H_{d,J}) = \int Q_y^0((E_J^0)_y) \nu_0(dy), \tag{A.7}$$

the integrand is measurable, and for every $y \in G_0$,

$$Q_y^0((E_J^0)_y) = Q_y^0(s_0(J; y)) \leq \widehat{K}_0(y) \lambda(s_0(J; y)). \tag{A.8}$$

For every interval J in either outer chart,

$$\mu(H_{d,J}) = \int Q_y^\infty((E_J^\infty)_y) \nu_\infty(dy), \tag{A.9}$$

the integrand is measurable, and for every $y \in G_\infty$,

$$Q_y^\infty((E_J^\infty)_y) = Q_y^\infty(s_\infty(J; y)) \leq \widehat{K}_\infty(y) \lambda(s_\infty(J; y)). \tag{A.10}$$

For $d \geq 2$ the two displays use their distinct conditioning kernels; for $d = 1$ they use the single unconditional kernel of α_0 .

Proof. We first prove parameterized measurability and disintegration in the exact product order used by the chart events. Fix a probability kernel Q_y from S to $\mathbb{R}$, where $S = \mathbb{R}^{d-1}$ or, for $d = 1$, $S = \mathbb{R}^0$. Put $W := \mathbb{R} \times S$ and

$$\mathcal{R} := \{A \times B : A \in \mathcal{B}(\mathbb{R}), B \in \mathcal{B}(S)\}.$$

This is a π -system, explicitly contains W , and generates the Borel sigma-algebra of W . Indeed, Borel rectangles are Borel, and every open subset of W is a countable union of open rectangles from countable bases of $\mathbb{R}$ and the Euclidean space S ; the one-point case is immediate.

Let $\mathcal{M}$ be the class of Borel sets $E \subseteq W$ for which $y \mapsto Q_y(E_y)$ is measurable. The whole product belongs to $\mathcal{M}$ because $Q_y(W_y) = Q_y(\mathbb{R}) = 1$. Rectangles belong to $\mathcal{M}$ because

$$Q_y((A \times B)_y) = \mathbf{1}_B(y)Q_y(A).$$

The class is closed under complements relative to W , since $Q_y((E^c)_y) = 1 - Q_y(E_y)$, and under countable disjoint unions, since kernel countable additivity gives

$$Q_y\left(\left(\bigcup_n E_n\right)_y\right) = \sum_n Q_y((E_n)_y),$$

whose partial sums are measurable and converge pointwise. Thus $\mathcal{M}$ is a λ -system containing $\mathcal{R}$, including its whole-space member. The π - λ theorem (Folland, 1999, Chapter 1) gives $\mathcal{M} = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(S)$.

Now suppose Q_y is a regular conditional law of X given Y , and let $\nu = \mathcal{L}(Y)$. Let $\mathcal{D}$ be the class of Borel $E \subseteq W$ for which

$$\mathbb{P}((X, Y) \in E) = \int Q_y(E_y) \nu(dy).$$

The preceding argument makes the integrand legal for every Borel E . The whole product belongs to $\mathcal{D}$, because both sides equal one. Every rectangle belongs to $\mathcal{D}$, since regular conditionality gives

$$\mathbb{P}((X, Y) \in A \times B) = \int_B Q_y(A) \nu(dy) = \int Q_y((A \times B)_y) \nu(dy).$$

Complements remain in $\mathcal{D}$ by subtraction from the equal whole-space mass. If (E_n) are pairwise disjoint members of $\mathcal{D}$, their sections are pairwise disjoint. Kernel countable additivity and monotone convergence (Folland, 1999, Chapter 2) give

$$\begin{aligned} \int Q_y\left(\left(\bigcup_n E_n\right)_y\right) \nu(dy) &= \int \sum_n Q_y((E_n)_y) \nu(dy) \\ &= \lim_{N \rightarrow \infty} \sum_{n=1}^N \int Q_y((E_n)_y) \nu(dy) \\ &= \sum_n \mathbb{P}((X, Y) \in E_n) \\ &= \mathbb{P}\left((X, Y) \in \bigcup_n E_n\right). \end{aligned}$$

Thus $\mathcal{D}$ is a λ -system containing $\mathcal{B}$, again with the whole product explicitly verified. A second π - λ argument extends the identity to every Borel $E \subseteq W$:

$$\mathbb{P}((X, Y) \in E) = \int Q_y(E_y) \nu(dy). \quad (\text{A.11})$$

Apply (A.11) to (X_0, Y_0) and the Borel set E_J^0 . Since $T_0(X_0, Y_0) = \alpha$, this proves (A.7). For $y \in G_0$, Lemma A.4 gives $(E_J^0)_y = [-R, R] \cap s_0(J; y)$, while Lemma A.3 gives $Q_y^0([-R, R]) = 1$. Intersecting with the cube therefore does not change conditional probability. The continuous image of an interval is an interval and hence Borel, so (A.1) applied to $s_0(J; y)$ proves (A.8).

Apply the same general identity separately to (X_∞, Y_∞) and E_J^∞ . The outer section formula, support of Q_y^∞ , and the same Borel-set domination prove (A.9) and (A.10). No inner kernel enters this calculation, and no outer kernel enters the inner calculation.

When $d = 1$, both coordinate pairs are $(\alpha_0, \emptyset)$, both reconstruction maps coincide, and the conditioning marginal is a unit mass. The kernel identification in Lemma A.3 then makes all four displays the correct unconditional identities. Empty chart pieces give empty images, and singleton pieces give singleton images; conditional absolute continuity assigns the latter probability zero. This completes the measurable conditional interface used below. $\square$

A.4 Inner, positive, and negative chart probability controls

Proposition A.4 (Inner chart probability control). *Under Assumptions 2, 3, and 4, and under Lemmas A.2 and A.3 and Proposition A.3, fix $\mu \in \mathcal{D}_{d,R,\eta}$ and $I \in \mathcal{I}(\Theta)$. With $I_0 = I \cap [-1, 1]$,*

$$\mu(H_{d,I_0}) \leq \bar{\kappa}_0 B_0(d, R) |I_0|.$$

If I_0 is empty or a singleton, then $\mu(H_{d,I_0}) = 0$.

Proof. Proposition A.3 gives a measurable integrand, and the full-measure set G_0 from Lemma A.3 permits us to write, without using a cap on a null fiber,

$$\begin{aligned} \mu(H_{d,I_0}) &= \int_{G_0} Q_y^0((E_{I_0}^0)_y) \nu_0(dy) \\ &= \int_{G_0} Q_y^0(s_0(I_0; y)) \nu_0(dy). \end{aligned}$$

For every $y \in G_0$, Lemma A.3 supplies finite $\widehat{K}_0(y)$ and simultaneous Borel-set domination, while Lemma A.2 supplies

$$\lambda(s_0(I_0; y)) \leq B_0(d, R) |I_0|.$$

Consequently, pointwise on G_0 ,

$$Q_y^0((E_{I_0}^0)_y) \leq \widehat{K}_0(y) \lambda(s_0(I_0; y)) \leq \widehat{K}_0(y) B_0(d, R) |I_0|.$$

The rightmost function is measurable because $\widehat{K}_0$ is measurable and $B_0(d, R) |I_0|$ is a fixed finite scalar. We therefore integrate only this measurable majorant, rather than requiring a separate measurability claim for the conditioning-dependent image length:

$$\begin{aligned} \mu(H_{d,I_0}) &\leq \int_{G_0} \widehat{K}_0(y) B_0(d, R) |I_0| \nu_0(dy) \\ &= B_0(d, R) |I_0| \int_{G_0} \widehat{K}_0(y) \nu_0(dy) \\ &\leq B_0(d, R) |I_0| \mathbb{E}_\mu \widehat{K}_0(Y_0) \\ &\leq \bar{\kappa}_0 B_0(d, R) |I_0|. \end{aligned}$$

If $I_0 = \emptyset$, the event is empty. If I_0 is a singleton, its image is a singleton and has Lebesgue measure zero; on G_0 the finite cap then gives conditional probability zero. Thus the degenerate conclusion does not involve a zero-times-infinity product, even when $\bar{\kappa}_0 = 0$. $\square$

Proposition A.5 (Positive outer chart probability control). *Under Assumptions 2, 3, and 4, and under Propositions A.1, A.3, and Lemma A.3, fix $\mu \in \mathcal{D}_{d,R,\eta}$ and $I \in \mathcal{I}(\Theta)$. With $I_+ = I \cap (1, \infty)$,*

$$\mu(H_{d,I_+}) \leq \bar{\kappa}_\infty B_\infty(d, R)|I_+|.$$

If I_+ is empty or a singleton, then $\mu(H_{d,I_+}) = 0$.

Proof. The positive-chart instance of Proposition A.3 and the full-measure property of G_∞ give

$$\begin{aligned} \mu(H_{d,I_+}) &= \int_{G_\infty} Q_y^\infty((E_{I_+}^\infty)_y) \nu_\infty(dy) \\ &= \int_{G_\infty} Q_y^\infty(s_\infty(I_+; y)) \nu_\infty(dy). \end{aligned}$$

For $d \geq 2$, $y \in [-R, R]^{d-1}$ on G_∞ ; appending the pivot value zero gives a support-compatible coefficient vector, so the deterministic outer certificate applies. For $d = 1$ the empty-tuple specialization applies directly. Thus Proposition A.1 gives

$$\lambda(s_\infty(I_+; y)) \leq B_\infty(d, R)|I_+|.$$

Combining this bound with finite-cap domination gives, pointwise on G_∞ ,

$$Q_y^\infty((E_{I_+}^\infty)_y) \leq \hat{K}_\infty(y) \lambda(s_\infty(I_+; y)) \leq \hat{K}_\infty(y) B_\infty(d, R)|I_+|.$$

The rightmost expression is a measurable cap times a fixed finite scalar. Integrating only that measurable majorant and then using the mean bound gives

$$\begin{aligned} \mu(H_{d,I_+}) &\leq \int_{G_\infty} \hat{K}_\infty(y) B_\infty(d, R)|I_+| \nu_\infty(dy) \\ &= B_\infty(d, R)|I_+| \int_{G_\infty} \hat{K}_\infty(y) \nu_\infty(dy) \\ &\leq B_\infty(d, R)|I_+| \mathbb{E}_\mu \hat{K}_\infty(Y_\infty) \\ &\leq \bar{\kappa}_\infty B_\infty(d, R)|I_+|. \end{aligned}$$

For an empty piece the event is empty. For a singleton, the image is a singleton and the finite-cap domination on G_∞ gives zero conditional probability. No conditioning-null fiber is used. $\square$

Proposition A.6 (Negative outer chart probability control). *Under Assumptions 2, 3, and 4, and under Propositions A.2, A.3, and Lemma A.3, fix $\mu \in \mathcal{D}_{d,R,\eta}$ and $I \in \mathcal{I}(\Theta)$. With $I_- = I \cap (-\infty, -1)$,*

$$\mu(H_{d,I_-}) \leq \bar{\kappa}_\infty B_\infty(d, R)|I_-|.$$

If I_- is empty or a singleton, then $\mu(H_{d,I_-}) = 0$.

Proof. The negative-chart instance of Proposition A.3 gives, after restriction to the finite-cap set,

$$\begin{aligned} \mu(H_{d,I_-}) &= \int_{G_\infty} Q_y^\infty((E_{I_-}^\infty)_y) \nu_\infty(dy) \\ &= \int_{G_\infty} Q_y^\infty(s_\infty(I_-; y)) \nu_\infty(dy). \end{aligned}$$

For $d \geq 2$, appending pivot value zero to the support-compatible y gives a coefficient vector in the cube; for $d = 1$, use the exact empty-tuple specialization. The specifically negative certificate gives

$$\lambda(s_\infty(I_-; y)) \leq B_\infty(d, R)|I_-|,$$

and finite-cap domination gives, pointwise on G_∞ ,

$$Q_y^\infty((E_{I_-}^\infty)_y) \leq \hat{K}_\infty(y)\lambda(s_\infty(I_-; y)) \leq \hat{K}_\infty(y)B_\infty(d, R)|I_-|.$$

The rightmost expression is measurable because it is the measurable cap times a fixed finite scalar. Integrating only this majorant gives

$$\begin{aligned} \mu(H_{d, I_-}) &\leq \int_{G_\infty} \hat{K}_\infty(y)B_\infty(d, R)|I_-|\nu_\infty(dy) \\ &= B_\infty(d, R)|I_-| \int_{G_\infty} \hat{K}_\infty(y)\nu_\infty(dy) \\ &\leq B_\infty(d, R)|I_-|\mathbb{E}_\mu \hat{K}_\infty(Y_\infty) \\ &\leq \bar{\kappa}_\infty B_\infty(d, R)|I_-|. \end{aligned}$$

An empty piece has an empty event, and a singleton has a singleton image of zero length; finite-cap domination makes its conditional probability zero. This is a separate negative-chart calculation and does not use a sign symmetry.

Together, the three propositions in this subsection give the three matching chart inequalities simultaneously. Each outer piece uses the mean outer cap once for its own event; no factor two has entered any individual estimate. When $d = 1$, Lemma A.3 identifies the inner and outer labels with the same unconditional coefficient law and the same finite density cap, while $s_0 = s_\infty = -\theta$ and $B_0(1, R) = B_\infty(1, R) = 1$. The inner proposition nevertheless invokes the primitive bound with $\bar{\kappa}_0$, and each outer proposition invokes the primitive bound with $\bar{\kappa}_\infty$. Thus coincidence of the degree-one kernel does not merge the two cap hypotheses or either higher-degree conditioning interface. $\square$

A.5 Three-piece decomposition and exact maximum

Lemma A.5 (Three-piece parameter and event decomposition). *Under Assumptions 1 and 2, for $I \in \mathcal{I}(\Theta)$ define I_0, I_+, I_- as in Section 2. Then, with every endpoint convention,*

$$I = I_0 \dot{\cup} I_+ \dot{\cup} I_-, \quad |I| = |I_0| + |I_+| + |I_-|,$$

the points 1 and -1 belong only to I_0 when they belong to I , and

$$H_{d, I} = H_{d, I_0} \cup H_{d, I_+} \cup H_{d, I_-}.$$

Proof. The sets $[-1, 1]$, $(1, \infty)$, and $(-\infty, -1)$ are pairwise disjoint and partition $\mathbb{R}$. Intersecting this exact partition with I preserves the endpoint convention and gives the disjoint union. In particular, an included 1 or -1 occurs only in I_0 ; an excluded boundary point occurs in none of the pieces. Finite additivity of Lebesgue measure gives the length identity, including empty and singleton pieces.

For a fixed coefficient vector, a root in I lies in exactly one of the three pieces, proving the forward event inclusion. A root in any piece is a root in I , proving the reverse inclusion. The three events need not be disjoint, since one polynomial can have roots in multiple pieces; the identity is an event union, not a probability-additivity assertion. $\square$

Proposition A.7 (Weighted chart union bound). *Under Assumptions 1 and 2, Lemma A.5, and Propositions A.4, A.5, and A.6, for every $\mu \in \mathcal{D}_{d,R,\eta}$ and $I \in \mathcal{I}(\Theta)$,*

$$\mu(H_{d,I}) \leq \bar{\kappa}_0 B_0(d, R)|I_0| + \bar{\kappa}_\infty B_\infty(d, R)(|I_+| + |I_-|).$$

If a piece is empty or a singleton, its event probability and weighted term are zero.

Proof. By Lemma A.5 and finite subadditivity,

$$\mu(H_{d,I}) \leq \mu(H_{d,I_0}) + \mu(H_{d,I_+}) + \mu(H_{d,I_-}).$$

Applying the three chart propositions gives

$$\begin{aligned} \mu(H_{d,I}) &\leq \bar{\kappa}_0 B_0(d, R)|I_0| + \bar{\kappa}_\infty B_\infty(d, R)|I_+| \\ &\quad + \bar{\kappa}_\infty B_\infty(d, R)|I_-|, \end{aligned}$$

which is the claimed factored form. Empty and singleton cases are already handled in the three preceding propositions and in the length identity; in particular, singleton boundary pieces at ± 1 require no correction. $\square$

Proposition A.8 (Exact two-chart maximum bound). *Under Assumptions 1 and 2, Lemma A.5, and Proposition A.7, every $\mu \in \mathcal{D}_{d,R,\eta}$ and $I \in \mathcal{I}(\Theta)$ satisfy*

$$\mu(H_{d,I}) \leq M_\eta(d, R)|I|, \quad M_\eta(d, R) = \max\{\bar{\kappa}_0 B_0(d, R), \bar{\kappa}_\infty B_\infty(d, R)\}.$$

Proof. For this scalar calculation set

$$A := \bar{\kappa}_0 B_0(d, R), \quad C := \bar{\kappa}_\infty B_\infty(d, R), \quad x := |I_0|, \quad y := |I_+|, \quad z := |I_-|.$$

The cap parameters are nonnegative and $d \geq 1$, $R \geq 1$, so $A, C, x, y, z \geq 0$. With $m = \max\{A, C\}$,

$$Ax + C(y + z) \leq mx + m(y + z) = \max\{A, C\}(x + y + z).$$

Lemma A.5 gives $x + y + z = |I|$, and Proposition A.7 gives the left-hand side. Since $\max\{A, C\} = M_\eta(d, R)$, the conclusion follows. This is the exact maximum of the chart coefficients; replacing it by $A + C$ would be a strictly weaker estimate and is not needed. $\square$

A.6 Class supremum and fixed-parameter polynomial specialization

Proposition A.9 (Class-supremum closure). *Under Assumption 2 and Proposition A.8, for every $d \geq 1$ and $R \geq 1$,*

$$C_{\mathcal{D}_{d,R,\eta}} \leq M_\eta(d, R).$$

This includes the convention that $C_{\mathcal{D}_{d,R,\eta}} = 0$ if either $\mathcal{D}_{d,R,\eta}$ or $\mathcal{I}(\Theta)$ is empty.

Proof. The quantities B_0 and B_∞ are nonnegative for the indexed regime, so $M_\eta(d, R) \geq 0$. If both indexing sets are nonempty, then for every admissible μ and every I the preceding proposition gives

$$\frac{\mu(H_{d,I})}{|I|} \leq M_\eta(d, R),$$

because $|I| > 0$. Taking first the interval supremum and then the law supremum preserves this inequality and yields the definition of $C_{\mathcal{D}_{d,R,\eta}}$. If either indexing set is empty, the stipulated value of $C_{\mathcal{D}_{d,R,\eta}}$ is zero, which is still at most the nonnegative $M_\eta(d, R)$. No witness or lower cap threshold is used. $\square$

Lemma A.6 (Inner coefficient domination). *Under Assumption 2, for every $d \geq 1$ and $R \geq 1$,*

$$\bar{\kappa}_0 B_0(d, R) \leq P_\eta(d, R).$$

Proof. By definition, $0 \leq \bar{\kappa}_0 \leq \bar{\kappa}_*$. Therefore

$$\begin{aligned} \bar{\kappa}_0 B_0(d, R) &= \bar{\kappa}_0 d + \frac{\bar{\kappa}_0}{2} R d(d-1) \\ &\leq \bar{\kappa}_* d + \frac{\bar{\kappa}_*}{2} R d(d-1) \\ &\leq \bar{\kappa}_* d + \frac{\bar{\kappa}_*}{2} R d^2 = P_\eta(d, R), \end{aligned}$$

where the second inequality is the pointwise comparison $d(d-1) \leq d^2$. All multipliers are nonnegative, including at $\bar{\kappa}_* = 0$. $\square$

Lemma A.7 (Outer coefficient domination). *Under Assumption 2, for every $d \geq 1$ and $R \geq 1$,*

$$\bar{\kappa}_\infty B_\infty(d, R) \leq P_\eta(d, R).$$

Proof. Since $0 \leq \bar{\kappa}_\infty \leq \bar{\kappa}_*$, $1 \leq d$, and $d(d-1) \leq d^2$, we have

$$\begin{aligned} \bar{\kappa}_\infty B_\infty(d, R) &= \bar{\kappa}_\infty + \frac{\bar{\kappa}_\infty}{2} R d(d-1) \\ &\leq \bar{\kappa}_* + \frac{\bar{\kappa}_*}{2} R d(d-1) \\ &\leq \bar{\kappa}_* d + \frac{\bar{\kappa}_*}{2} R d^2 = P_\eta(d, R). \end{aligned}$$

The inequality remains valid at the zero-cap boundary. $\square$

Proposition A.10 (Fixed- η polynomial specialization). *Under Assumption 2 and Lemmas A.6 and A.7, for every $d \geq 1$ and $R \geq 1$,*

$$M_\eta(d, R) \leq P_\eta(d, R) = \bar{\kappa}_* d + \frac{\bar{\kappa}_*}{2} R d^2.$$

Proof. The two preceding lemmas bound both entries of the maximum defining $M_\eta(d, R)$ by the same $P_\eta(d, R)$. Maximum monotonicity therefore gives

$$\max\{\bar{\kappa}_0 B_0(d, R), \bar{\kappa}_\infty B_\infty(d, R)\} \leq \max\{P_\eta(d, R), P_\eta(d, R)\} = P_\eta(d, R).$$

This is pointwise for every allowed index, not asymptotic. For fixed η , P_η has monomials d and Rd^2 , total degree at most three in (d, R) , and its displayed coefficients depend only on η . If $\bar{\kappa}_* = 0$, it is the zero polynomial and the same comparison holds. The proof also verifies every technical condition used by the public fixed-cap corollary: $d \geq 1$, $R \geq 1$, nonnegative cap parameters, and the two explicit inequalities $1 \leq d$ and $d(d-1) \leq d^2$. The passage from the per-law inequality to the class constant is deterministic and was already proved in Proposition A.9; no probability, horizon, or norm conversion is hidden in this specialization. $\square$

A.7 Witness kernels, support, dependence, and membership

Proposition A.11 (Degree-one endpoint kernels). *Under Assumption 2, if $d = 1$ and $\mu_{1,R}^{\text{wit}}$ is the witness law defined above, then both regular conditional laws in the definition of $\mathcal{D}_{1,R,\eta}$ are the marginal law*

$$\nu_R(A) := \frac{\lambda(A \cap [-R, R])}{2R} = \int_A u_R(x) dx, \quad u_R(x) := \frac{1}{2R} \mathbf{1}_{[-R, R]}(x).$$

Consequently,

$$K_0^{\mu_{1,R}^{\text{wit}}} = K_\infty^{\mu_{1,R}^{\text{wit}}} = \frac{1}{2R} \quad \text{almost surely.}$$

Proof. For $d = 1$, the sole coefficient is uniform on $[-R, R]$, so its law and density are exactly ν_R and u_R . Both conditioning tuples are empty; equivalently, the conditioning space is the one-point space $\mathbb{R}^0$, and the conditional law at that point is the marginal ν_R . Since $R \geq 1$, $[-R, R]$ has positive length and

$$\|u_R\|_{L^\infty(\mathbb{R})} = \frac{1}{2R}.$$

Every other density version agrees with u_R almost everywhere and has the same essential supremum. Both setting-defined caps therefore equal the displayed deterministic value. $\square$

Proposition A.12 (Degree-two full-complement kernels). *Under Assumption 2, if $d = 2$ and $\mu_{2,R}^{\text{wit}}$ is the witness law, then a regular conditional law of α_0 given α_1 and a regular conditional law of α_1 given α_0 are both the constant kernel ν_R , whose density is $u_R = (2R)^{-1} \mathbf{1}_{[-R, R]}$. Consequently,*

$$K_0^{\mu_{2,R}^{\text{wit}}} = K_\infty^{\mu_{2,R}^{\text{wit}}} = \frac{1}{2R} \quad \text{almost surely.}$$

Proof. Here α_0 and α_1 are independent with common law ν_R . For every Borel $A \subseteq \mathbb{R}$, define the constant kernels

$$Q_y^0(A) := \nu_R(A), \quad Q_x^\infty(A) := \nu_R(A) \quad (x, y \in \mathbb{R}).$$

They are measurable in their conditioning variables. For Borel $A, C \subseteq \mathbb{R}$, independence gives

$$\begin{aligned} \mathbb{P}(\alpha_0 \in A, \alpha_1 \in C) &= \nu_R(A) \mathbb{P}(\alpha_1 \in C) = \int_C Q_y^0(A) \mathbb{P}_{\alpha_1}(dy), \\ \mathbb{P}(\alpha_1 \in A, \alpha_0 \in C) &= \nu_R(A) \mathbb{P}(\alpha_0 \in C) = \int_C Q_x^\infty(A) \mathbb{P}_{\alpha_0}(dx). \end{aligned}$$

Thus they are regular conditional laws after conditioning on the full nonpivot coordinate. Each has density u_R and essential supremum $1/(2R)$. The integral identities determine regular conditional laws on a full conditioning-marginal set (Kallenberg, 2002, Theorem 6.3); Radon–Nikodym uniqueness (Folland, 1999, Theorem 3.8) then makes the density norm independent of the permissible versions. Hence both random caps have the asserted value almost surely. $\square$

Proposition A.13 (Higher-degree full-complement kernels). *Under Assumption 2, if $d \geq 3$ and $\mu_{d,R}^{\text{wit}}$ is the witness law, then a regular conditional law of α_0 given all coordinates $\alpha_{1:d-1}$ and a regular conditional law of α_{d-1} given all coordinates $\alpha_{0:d-2}$ are both the constant uniform kernel ν_R , with density $u_R = (2R)^{-1} \mathbf{1}_{[-R, R]}$. Consequently,*

$$K_0^{\mu_{d,R}^{\text{wit}}} = K_\infty^{\mu_{d,R}^{\text{wit}}} = \frac{1}{2R} \quad \text{almost surely.}$$

Proof. The construction is

$$\alpha_0 = U_0, \quad \alpha_j = RS \ (1 \leq j \leq d-2), \quad \alpha_{d-1} = U_\infty,$$

where $U_0, U_\infty \sim \nu_R$, S is Rademacher, and all three variables are mutually independent. The complete conditioning tuples are

$$\alpha_{1:d-1} = (RS, \dots, RS, U_\infty), \quad \alpha_{0:d-2} = (U_0, RS, \dots, RS).$$

The first is a measurable function of (S, U_∞) and is independent of $U_0 = \alpha_0$. The second is a measurable function of (U_0, S) and is independent of $U_\infty = \alpha_{d-1}$. Although either complement reveals S on the witness support, it does not reveal the corresponding endpoint pivot.

For all $y, z \in \mathbb{R}^{d-1}$, define

$$Q_y^0(A) := \nu_R(A), \quad Q_z^\infty(A) := \nu_R(A).$$

For Borel $C \subseteq \mathbb{R}^{d-1}$, the two independence statements give

$$\begin{aligned} \mathbb{P}(\alpha_0 \in A, \alpha_{1:d-1} \in C) &= \nu_R(A) \mathbb{P}(\alpha_{1:d-1} \in C) \\ &= \int_C Q_y^0(A) \mathbb{P}_{\alpha_{1:d-1}}(dy), \end{aligned}$$

and

$$\mathbb{P}(\alpha_{d-1} \in A, \alpha_{0:d-2} \in C) = \int_C Q_z^\infty(A) \mathbb{P}_{\alpha_{0:d-2}}(dz).$$

The constant kernels are measurable, so these identities verify them as regular conditional laws on the full complement spaces, even at conditioning values outside the witness support. Their common density is u_R , whose essential supremum is $1/(2R)$. Almost-sure uniqueness of conditional laws (Kallenberg, 2002, Theorem 6.3) and of Radon–Nikodym densities (Folland, 1999, Theorem 3.8) shows that every permissible version has the same realized cap. Thus the two caps are exactly, rather than merely at most, $1/(2R)$. $\square$

Proposition A.14 (Cube support and ambient singularity). *Under Assumption 2, the witness satisfies*

$$\mu_{d,R}^{\text{wit}}([-R, R]^d) = 1 \quad \text{for every } d \geq 1.$$

For every $d \geq 3$, it is singular with respect to d -dimensional Lebesgue measure.

Proof. For $d = 1$, the only coordinate is uniform on $[-R, R]$. For $d = 2$, both coordinates are uniform on this interval. For $d \geq 3$, the two endpoints lie in $[-R, R]$ almost surely and every middle coordinate is $RS \in \{-R, R\}$. This proves cube support in all three regimes.

For $d \geq 3$, set

$$A_{d,R} := [-R, R] \times \{(R, \dots, R), (-R, \dots, -R)\} \times [-R, R] \subseteq \mathbb{R}^d,$$

where each middle vector has $d-2 \geq 1$ entries. This is a closed Borel set and $\mu_{d,R}^{\text{wit}}(A_{d,R}) = 1$. Its middle factor is a finite subset of $\mathbb{R}^{d-2}$ and has $(d-2)$ -dimensional Lebesgue measure zero. Product measure on the two product rectangles comprising $A_{d,R}$ (Folland, 1999, Theorem 2.37) gives

$$\lambda_d(A_{d,R}) = 0.$$

Thus a Lebesgue-null Borel set carries all witness mass. The argument includes $d = 3$, where the middle factor is the two-point set $\{-R, R\}$. $\square$

Proposition A.15 (Dependence of the middle coordinates). *Under Assumption 2, if $d \geq 4$, then the middle coordinates of $\mu_{d,R}^{\text{wit}}$ are dependent; in particular, α_1 and α_2 are not independent.*

Proof. Both coordinates exist when $d \geq 4$, and $\alpha_1 = RS = \alpha_2$ almost surely. Since $R \geq 1$, the values R and $-R$ are distinct. Hence

$$\mathbb{P}(\alpha_1 = R) = \mathbb{P}(\alpha_2 = R) = \frac{1}{2}, \quad \mathbb{P}(\alpha_1 = R, \alpha_2 = R) = \frac{1}{2}.$$

The joint probability is not the product $1/4$ of the marginals, proving dependence. $\square$

Lemma A.8 (Witness-cap threshold comparison). *Under Assumption 2, suppose that the applicable one of Propositions A.11, A.12, and A.13 gives*

$$K_0^{\mu_{d,R}^{\text{wit}}} = K_\infty^{\mu_{d,R}^{\text{wit}}} = \frac{1}{2R} \quad \text{almost surely.}$$

If, only for this witness clause, $\bar{\kappa}_0, \bar{\kappa}_\infty \geq 1/2$, then

$$\mathbb{E}K_0^{\mu_{d,R}^{\text{wit}}} \leq \bar{\kappa}_0, \quad \mathbb{E}K_\infty^{\mu_{d,R}^{\text{wit}}} \leq \bar{\kappa}_\infty.$$

Proof. Each cap is the deterministic value $1/(2R)$. Since $R \geq 1$,

$$\mathbb{E}K_0^{\mu_{d,R}^{\text{wit}}} = \mathbb{E}K_\infty^{\mu_{d,R}^{\text{wit}}} = \frac{1}{2R} \leq \frac{1}{2} \leq \min\{\bar{\kappa}_0, \bar{\kappa}_\infty\}.$$

This is the only use of either lower threshold on the cap parameters. $\square$

Proposition A.16 (Witness membership). *Under Assumption 2 and Propositions A.11–A.15 and Lemma A.8, if $\bar{\kappa}_0, \bar{\kappa}_\infty \geq 1/2$, then for every $d \geq 1$ and $R \geq 1$,*

$$\mu_{d,R}^{\text{wit}} \in \mathcal{D}_{d,R,\eta}, \quad K_0^{\mu_{d,R}^{\text{wit}}} = K_\infty^{\mu_{d,R}^{\text{wit}}} = \frac{1}{2R} \quad \text{almost surely.}$$

The witness is ambiently singular for $d \geq 3$ and has dependent middle coordinates for $d \geq 4$.

Proof. Fix (d, R) and impose the lower thresholds only for this membership statement.

If $d = 1$, Proposition A.11 supplies the two conditional densities and exact caps for the empty conditioning tuples. Proposition A.14 supplies support on $[-R, R]$, and Lemma A.8 supplies both mean-cap inequalities.

If $d = 2$, Proposition A.12 supplies the two full-complement conditional densities and caps, now conditioning each endpoint on the other. Proposition A.14 gives support on $[-R, R]^2$, and Lemma A.8 supplies the mean inequalities.

If $d \geq 3$, Proposition A.13 supplies both full-complement conditional densities and caps even though the complement contains the Rademacher middle block. Proposition A.14 gives cube support and ambient singularity. When $d \geq 4$, Proposition A.15 additionally proves the middle-coordinate dependence. Lemma A.8 again gives the two mean-cap inequalities.

In every degree regime, the witness is the law of a Borel measurable Euclidean random vector and hence is a Borel probability law on $\mathbb{R}^d$. The preceding results verify every item in the definition of $\mathcal{D}_{d,R,\eta}$: cube support, almost-sure existence of both endpoint conditional densities, and both mean-cap inequalities. This proves membership and the exact cap identities. Singularity and middle dependence show that neither an ambient joint density nor coordinate independence has been used as a hidden membership condition. The threshold comparison is isolated in Lemma A.8 and has no role in the general root-hitting argument. $\square$

A.8 Proof of the Main Theorem

Proof of Theorem 3.1. Fix the finite η first, and then fix arbitrary $d \geq 1$, $R \geq 1$, $\mu \in \mathcal{D}_{d,R,\eta}$, and $I \in \mathcal{I}(\Theta)$. Proposition A.7 gives the exact weighted three-piece estimate

$$\mu(H_{d,I}) \leq \bar{\kappa}_0 B_0(d, R) |I_0| + \bar{\kappa}_\infty B_\infty(d, R) (|I_+| + |I_-|).$$

Proposition A.8 then gives

$$\mu(H_{d,I}) \leq M_\eta(d, R) |I|.$$

The first proposition uses only the finite union bound after the exact event decomposition, and the second uses the scalar maximum inequality, so the coefficient is the maximum of the two chart constants rather than their sum.

Proposition A.9 divides by the positive interval length and takes the interval and law suprema, while also treating either empty indexing set. Hence

$$C_{\mathcal{D}_{d,R,\eta}} \leq M_\eta(d, R).$$

Proposition A.10 proves, pointwise for every allowed d, R ,

$$M_\eta(d, R) \leq P_\eta(d, R) = \bar{\kappa}_* d + \frac{\bar{\kappa}_*}{2} R d^2.$$

This completes the root-hitting and class-uniform conclusions without any lower bound on $\bar{\kappa}_0$ or $\bar{\kappa}_\infty$.

Under the separate hypothesis $\bar{\kappa}_0, \bar{\kappa}_\infty \geq 1/2$, Proposition A.16 gives, simultaneously for every $d \geq 1$ and $R \geq 1$,

$$\mu_{d,R}^{\text{wit}} \in \mathcal{D}_{d,R,\eta}, \quad K_0^{\mu_{d,R}^{\text{wit}}} = K_\infty^{\mu_{d,R}^{\text{wit}}} = \frac{1}{2R} \quad \text{almost surely.}$$

This proves the independent nonemptiness clause.

Every constant used above is one of the displayed setting-derived $B_0, B_\infty, M_\eta, P_\eta$. The chart estimates are per-law deterministic probability inequalities, and the remaining operations are deterministic finite addition, scalar domination, and suprema. Thus no confidence parameter, horizon upgrade, limiting argument, norm conversion, or hidden dependence is introduced. The exposed quantities and the static Lebesgue-length and conditional- L^∞ modes are exactly those stated in Theorem 3.1. $\square$

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